

6.0.1 UGC NET CSE | December 2004 | Part 2 | Question: 49



A data warehouse is always _____.

- A. Subject oriented B. Object oriented
C. Program oriented D. Compiler oriented

ugcnetcse-dec2004-paper2

Answer key

6.1

Data Cube (1)

6.1.1 Data Cube: UGC NET CSE | December 2015 | Part 2 | Question: 31



A data cube C , has n dimensions, and each dimensions has exactly p distinct values in the base cuboid. Assume that there are no concept hierarchies associated with the dimensions. What is the maximum number of cells possible in the data cube, C ?

- A. p^n B. p C. $(2^n - 1)(p + 1)$ D. $(p + 1)^n$

ugcnetcse-dec2015-paper2 data-cube data-modelling

Answer key

6.2

Data Mining (1)

6.2.1 Data Mining: UGC NET CSE | October 2020 | Part 2 | Question: 35



The process of removing details from a given state representation is called _____

- A. Extraction B. Mining C. Selection D. Abstraction

ugcnetcse-oct2020-paper2 non-gatecse data-mining

Answer key

6.3

Data Mining and Warehousing (1)

6.3.1 Data Mining and Warehousing: UGC NET CSE | December 2015 | Part 2 | Question: 34



_____ allows selection of the relevant information necessary for the data warehouse:

- A. The Top-Down View B. Data Warehouse View
C. Datasource View D. Business Query View

ugcnetcse-dec2015-paper2 data-warehousing data-mining-and-warehousing

Answer key

6.4

Decision Tree (1)

6.4.1 Decision Tree: UGC NET CSE | September 2013 | Part 2 | Question: 40



The decision tree classifier is a widely used technique for

- A. Classification B. Association C. Partition D. Clustering

ugcnetsep2013iii machine-learning decision-tree

Answer key

6.5

Expert System (1)

6.5.1 Expert System: UGC NET CSE | December 2014 | Part 3 | Question: 59



Match the following components of an expert system :

List – I

- a. I/O Interface
- b. Explanation Module
- c. Inference Engine
- d. Knowledge Base

List – II

- i. Accepts user's queries and responds to question through I/O interface
- ii. Contains facts and rules about the domain
- iii. Gives the user, the ability to follow inferencing steps at any time during consultation
- iv. Permits the user to communicate with the system in a natural way

Codes :

- A. a-i, b-iii, c-iv, d-ii
- C. a-i, b-iii, c-ii, d-iv

- B. a-iv, b-iii, c-i, d-ii
- D. a-iv, b-i, c-iii, d-ii

ugcnetcse-dec2014-paper3 artificial-intelligence expert-system

Answer key

6.6

Fuzzy Set (1)

6.6.1 Fuzzy Set: UGC NET CSE | June 2012 | Part 3 | Question: 6



If two fuzzy set A and B are given with membership functions

$$\mu_A(x) = \{0.2, 0.4, 0.8, 0.5, 0.1\}$$

$$\mu_B(x) = \{0.1, 0.3, 0.6, 0.3, 0.2\}$$

Then the value of $\mu_{\overline{A \cap B}}$ will be

- A. $\{0.9, 0.7, 0.4, 0.8, 0.9\}$
- C. $\{0.1, 0.3, 0.6, 0.3, 0.1\}$

- B. $\{0.2, 0.4, 0.8, 0.5, 0.2\}$
- D. $\{0.7, 0.3, 0.4, 0.2, 0.7\}$

ugcnetcse-june2012-paper3 fuzzy-set

Answer key

6.7

Group Decision Support System (1)

6.7.1 Group Decision Support System: UGC NET CSE | August 2016 | Part 3 | Question: 73



Criticism free idea generation is a factor of _____.

- A. Decision Support System
- C. Enterprise Resource Support System

- B. Group Decision Support System
- D. Artificial Intelligence

ugcnetcse-aug2016-paper3 artificial-intelligence group-decision-support-system

Answer key

6.8

K Means Clustering (1)

6.8.1 K Means Clustering: UGC NET CSE | October 2020 | Part 2 | Question: 83



Given below are two statements:

If two variables V_1 and V_2 are used for clustering, then consider the following statements for k means clustering with $k = 3$

- Statement I: If V_1 and V_2 have correlation of 1 the cluster centroid will be in straight line
- Statement II: If V_1 and V_2 have correlation of 0 the cluster centroid will be in straight line

In the light of the above statements, choose the correct answer from the options given below

- A. Both Statement I and Statement II are true
- B. Both Statement I and Statement II are false

- C. Statement I is correct but Statement II is false
D. Statement I is incorrect but Statement II is true

ugcnetcse-oct2020-paper2 k-means-clustering

Answer key 

6.9 Knowledge Representation (5)

6.9.1 Knowledge Representation: UGC NET CSE | December 2013 | Part 3 | Question: 23



High level knowledge which relates to the use of sentences in different contexts and how the context affect the meaning of the sentences?

- A. Morphological B. Syntactic C. Semantic D. Pragmatic

ugcnetcse-dec2013-paper3 artificial-intelligence knowledge-representation

Answer key 

6.9.2 Knowledge Representation: UGC NET CSE | December 2014 | Part 3 | Question: 58



Match the following :

List – I

- a. Script
b. Conceptual Dependencies
c. Frames
d. Associative Network

List – II

- i. Directed graph with labelled nodes for graphical representation of knowledge
ii. Knowledge about objects and events is stored in record-like structures consisting of slots and slot values.
iii. Primitive concepts and rules to represent natural language statements
iv. Frame like structures used to represent stereotypical patterns for commonly occurring events in terms of actors, roles, props and scenes.

Codes :

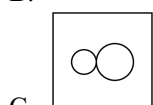
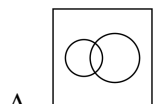
- A. a-iv, b-ii, c-i, d-iii B. a-iv, b-iii, c-ii, d-i
C. a-ii, b-iii, c-iv, d-i D. a-i, b-iii, c-iv, d-ii

ugcnetcse-dec2014-paper3 artificial-intelligence knowledge-representation

6.9.3 Knowledge Representation: UGC NET CSE | June 2010 | Part 2 | Question: 2



' R is a robot of M ' means R can perform some of the tasks that otherwise M would do and R is unable to do anything else. Which of the following is the most appropriate representation to model this situation?



D. None of these

ugcnetcse-june2010-paper2 knowledge-representation non-gatecse

6.9.4 Knowledge Representation: UGC NET CSE | June 2013 | Part 3 | Question: 72

Which of the following is a knowledge representation technique used to represent knowledge about stereotype situation?

- A. Semantic network B. Frames C. Scripts D. Conceptual Dependency

ugcnetcse-june2013-paper3 artificial-intelligence knowledge-representation

Answer key

6.9.5 Knowledge Representation: UGC NET CSE | June 2014 | Part 3 | Question: 08

Consider a fuzzy set old as defined below

Old = $\{(20, 0.1), (30, 0.2), (40, 0.4), (50, 0.6), (60, 0.8), (70, 1), (80, 1)\}$.

Then the alpha-cut for $\alpha = 0.4$ for the set old will be

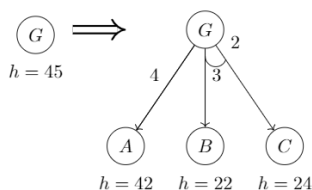
- A. $\{(40, 0.4)\}$
 B. $\{50, 60, 70, 80\}$
 C. $\{(20, 0.1), (30, 0.2)\}$
 D. $\{(20, 0), (30, 0), (40, 1), (50, 1), (60, 1), (70, 1), (80, 1)\}$

ugcnetjune2014iii knowledge-representation fuzzy-set

Answer key

6.10 Non Gatecse (1)**6.10.1 Non Gatecse: UGC NET CSE | January 2017 | Part 3 | Question: 56**

Consider the following AO graph:



Which is the best node to expand by next AO* algorithm?

- A. A B. B C. C D. B and C

ugcnetcse-jan2017-paper3 non-gatecse

Answer key

6.11 Olap (1)**6.11.1 Olap: UGC NET CSE | December 2015 | Part 2 | Question: 33**

_____ system is market oriented and is used for data analysis by knowledge workers including Managers, Executives and Analysts.

- A. OLTP B. OLAP C. Data System D. Market System

ugcnetcse-dec2015-paper2 data-warehousing olap data-mining-and-warehousing

Answer key

6.12 Perceptron (1)**6.12.1 Perceptron: UGC NET CSE | December 2014 | Part 3 | Question: 70**

Consider the following statements about a perception :

I. Feature detector can be any function of the input parameters.

II. Learning procedure only adjusts the connection weights to the output layer. Identify the correct statement out of the following :

A. *I* is false and *II* is false.

B. *I* is true and *II* is false.

C. *I* is false and *II* is true.

D. *I* is true and *II* is true.

ugcnetcse-dec2014-paper3 neural-network perceptron

Answer key 

Answer Keys

6.0.1	Q-Q
6.5.1	B
6.9.2	B
6.11.1	B

6.1.1	D
6.6.1	X
6.9.3	Q-Q
6.12.1	D

6.2.1	D
6.7.1	Q-Q
6.9.4	C

6.3.1	A
6.8.1	C
6.9.5	D

6.4.1	A
6.9.1	D
6.10.1	A